

## 4.2 Random Variables and Their Properties

### Experiment, Sample Space, and Sample Points

An **experiment** is any process where the result is **not known for sure in advance**.

- The **sample space (S)** is the **set of all possible outcomes** of the experiment.
- Each individual outcome in the sample space is called a **sample point**.

#### Example 4.1 (Coin Toss)

If we flip a coin:  $S = \{H, T\}$

Here, **H = Head** and **T = Tail**.

The symbol  $\{ \}$  means “the set containing”.

#### Example 4.2 (Throwing a Die)

If we throw a die:  $S = \{1, 2, 3, 4, 5, 6\}$

Each number represents the face that appears on the die.

### Random Variable

A **random variable** is a **rule or function** that assigns a **real number** to each outcome in the sample space.

- Random variables are usually written using **capital letters** like **X, Y, Z**.
- The values they take are written using **small letters** like **x, y, z**.

#### Example 4.3 (Two Dice)

If we roll two dice, the sample space is:

$S = \{(1,1), (1,2), \dots, (6,6)\}$

Each pair  $((i, j))$  shows the result of the **first and second die**. If  $X = \text{sum of the two dice}$ , then:

- For outcome  $(4, 3)$ ,  **$X = 7$**

So, the random variable converts outcomes into numbers.

#### Example 4.4 (Two Coins)

If we flip two coins and  **$X = \text{number of heads}$** :

- $(H, T) \rightarrow X = 1$
- $(T, H) \rightarrow X = 1$

Different outcomes can give the **same value** of a random variable.

## Distribution Function (Cumulative Distribution Function – CDF)

The **distribution function** of a random variable **X**, written as **F(x)**, is defined as:

$$F(x) = P(X \leq x)$$

This means:

- **F(x)** is the **probability that X takes a value less than or equal to x**.
- Probability starts from **0** and eventually reaches **1**.

## Discrete Random Variables

A random variable is **discrete** if it can take **only a finite or countable number of values**.

Examples:

- Number of heads in coin tosses
- Demand size of a product
- Number of defective items

If **X** is discrete and takes values  $(x_1, x_2, x_3, \dots)$ , then:

$$p(x_i) = P(X = x_i)$$

This is called the **probability mass function (PMF)**.

The total probability must satisfy:

$$\sum p(x_i) = 1$$

### Example 4.6 (Quality Inspection)

A manufactured part can be:

- **Good (1)** with probability **0.9**
- **Bad (0)** with probability **0.1**

So, **X** is a **discrete random variable** with:

$$p(0) = 0.1, p(1) = 0.9$$

This is an example of a Bernoulli random variable.

## Continuous Random Variables

A random variable **X** is **continuous** if it can take **infinitely many values** (for example, time, height, weight).

For a continuous random variable, there exists a **probability density function (PDF)** ( $f(x)$ ) such that:  $P(X \in B) = \int_B f(x) dx$

Also:

$$\left[ \int_{-\infty}^{\infty} f(x) dx = 1 \right]$$

This means:

- The **total area under the curve  $f(x)$**  is **1**
- Probabilities are found by **areas**, not by single points

All probability statements for a continuous random variable can be obtained from  **$f(x)$** .

## Discrete vs Continuous Random Variables: Meaning of $p(x)$ and $f(x)$

### Discrete Case

For a **discrete random variable**,  **$p(x)$**  is the **actual probability** that the random variable  **$X$**  equals  **$x$** .

Example:

If  $X$  = number of heads in one coin toss, then

$$p(1) = P(X=1)$$

So for discrete variables:

- Probabilities are **directly assigned to values**
- Each value can have a **non-zero probability**

### Continuous Case

For a **continuous random variable**, things are different.

- **$f(x)$  is NOT the probability that  $X = x$**
- In fact,  $P(X = x) = 0$ . Why?

Because a continuous random variable can take **infinitely many values**, so the probability of any **single exact value** is zero.

$$\text{Mathematically } P(X = x) = P(X \in [x, x]) = \int_x^x f(y) dy = 0$$

## Meaning of the Probability Density Function $f(x)$

Although  **$f(x)$  is not a probability**, it helps us find probabilities over **intervals**.

If ( $\Delta x > 0$ ), then:

$$P(X \in [x, x + \Delta x]) = \int_x^{x + \Delta x} f(y) dy$$

## Geometric Interpretation

- This probability is the **area under the curve  $f(x)$**
- Between  **$x$  and  $x + \Delta x$**
- (Shown in Fig. 4.3)

A continuous random variable is:

- **More likely** to fall in regions where  **$f(x)$  is large**
- **Less likely** where  **$f(x)$  is small**

## Distribution Function for Continuous Random Variables

The **distribution function (CDF)** is defined as:  $F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$   
This means:

- **$F(x)$**  gives the probability that  **$X$  is less than or equal to  $x$**

### Relationship between $F(x)$ and $f(x)$

Under normal conditions:  $f(x) = F'(x)$  So:

- **$f(x)$**  is the **derivative** of  **$F(x)$**
- **$F(x)$**  is the **integral** of  **$f(x)$**

### Probability Over an Interval

If (  $I = [a,b]$  ), then:

$$P(X \in I) = \int_a^b f(y) dy = F(b) - F(a)$$

This result comes from the Fundamental Theorem of Calculus.

## Example 4.7: Uniform Random Variable on $[0,1]$

All intervals of the **same length** have **equal probability**

That's why it's called **uniform**

### Importance in Simulation

The uniform random variable on  $[0,1]$  is **very important**, because:

- It is used to **generate all other random variables on a computer**

## Example 4.8: Exponential Random Variable

The **exponential distribution** is commonly used for:

- Interarrival times
- Service times
- Demand times

It has: A PDF ( $f(x)$ ) A CDF ( $F(x)$ )  
(both shown in the figures)

## Multiple Random Variables

In simulations, we usually work with **many random variables together**, for example:

- Service times ( $S_1, S_2, \dots$ )
- Waiting times ( $D_1, D_2, \dots$ )

For simplicity, the text explains the case of **two random variables: X and Y**.

## Jointly Discrete Random Variables

### Example 4.9

- **X and Y are independent**((( $\hat{\theta}$ )))

## Jointly Continuous Random Variables

X and Y are **jointly continuous** if there exists a function  $f(x,y)$  such that:

### Independence Condition

$$f(x,y) = f_X(x)f_Y(y)$$

### Example 4.11

After computing marginals:

- **X and Y are dependent**

## Intuitive Meaning of Independence

Two random variables are **independent** if:

- Knowing the value of one **gives no information** about the other

If this is not true:

- They are **dependent**

## Mean (Expected Value)

The **mean** or **expected value** of a random variable ( $X_i$ ) is denoted by.

Interpretation: The mean represents the **center of gravity** of the distribution and is a **measure of central tendency**.

## Median: Another Measure of Central Tendency

The **median  $x_{0.5}$**  is defined as:

- The **smallest value of  $x$**  such that:

$$F_X(x) \geq 0.5$$

$$F_X(x) \geq 0.5$$

For a **continuous random variable**:

$$F_X(x_{\{0.5\}}) = 0.5$$

$$F_X(x_{0.5}) = 0.5$$

### Why Median is Important

- The median is **not affected by extreme values**
- It can be a **better measure than the mean** when very large or very small values exist

### Example 4.14

- Random variable **X** takes values: 1,2,3,4,5 (each with probability 0.2)
  - Mean = 3
  - Median = 3
- Random variable **Y** takes values: 1,2,3,4,100 (each with probability 0.2)
  - Mean = 22
  - Median = 3

The **mean changes a lot**, but the **median stays the same**.  
So, the median is **insensitive to extreme values**.

## Mode

The **mode** is the value of **x** that:

- Maximizes **p(x)** for discrete variables
- Maximizes **f(x)** for continuous variables

The mode is the **most likely value** of the random variable.  
A distribution can have **more than one mode**.

## What is a Stochastic Process?

A **stochastic process** is a **collection of similar random variables ordered over time**, all defined on the same probability space.

- The set of all possible values the variables can take is called the **state space**.

### Types of Stochastic Processes

**1. Discrete-time stochastic process**

If the random variables are indexed by integers:  $X_1, X_2, X_3, \dots$

**2. Continuous-time stochastic process**

If the random variables are defined for all time values:  $\{X(t), t \geq 0\}$

## Example 4.19: Queueing System (M/M/1 Queue)

Consider a **single-server queue**:

- Interarrival times:  $A_1, A_2, \dots$  (IID)
- Service times:  $S_1, S_2, \dots$  (IID)
- Service rule: **FIFO** (First-In-First-Out)

Let:  $D_i$  = delay of the  $i$ -th customer in the queue

The delay process is defined as:

$$D_1 = 0$$

- $D_{i+1} = \max\{D_i + S_i - A_{i+1}, 0\}$

### Important Points

- The simulation **converts input random variables** (arrival and service times) into an **output stochastic process**  $D_1, D_2, \dots$
- The **state space** is all **non-negative real numbers**
- $D_i$  and  $D_{i+1}$  are **positively correlated**  
(because if one customer waits a lot, the next one is also likely to wait)

## Example 4.20: Number in Queue Over Time

Let:

- $Q(t)$  = number of customers in the queue at time  $t$

Then:

- $\{Q(t), t \geq 0\}$  is a **continuous-time stochastic process**
- State space =  $\{0, 1, 2, \dots\}$

## Example 4.21: Inventory System

Let:

- $C_i$  = total cost in month  $i$  (ordering + holding + shortage)

Then:

- $C_1, C_2, \dots$  is a **discrete-time stochastic process**
- State space = **non-negative real numbers**

## Why Assumptions Are Needed

To analyze simulation output statistically, we often **assume certain properties** about the stochastic process.

Without such assumptions, **statistical analysis becomes impossible**.

One very important assumption is **covariance stationarity**.

## Key Observations from the Diagram for 4.22((XM))

### All correlations are positive

$r_j > 0$  for all  $j$

**Meaning:**

If one customer experiences a large delay, the next customers are **also likely to experience large delays**. This happens because:

- Heavy traffic creates congestion
- Congestion does not disappear instantly

So delays move **together**, not independently.

### Correlation decreases as lag increases

$r_1 > r_2 > r_3 > \dots$

**Meaning:**

- $D_i$  and  $D_{i+1}$  are strongly related
- $D_i$  and  $D_{i+10}$  are weakly related

**Why?**

The effect of congestion fades as time passes.

### High traffic $\rightarrow$ slow decay of correlation

When traffic intensity is **high** (e.g.,  $\rho = 0.9$ ):



- Correlation remains **large even for big lags**
- Curve decreases **very slowly**

#### Interpretation:

When the system is almost overloaded:

- Long queues persist
- Delays stay high for many customers Output data are **strongly dependent**

➡ Low traffic → fast decay of correlation

When traffic intensity is **low** (e.g.,

$\rho=0.5$ ):

- Correlation drops quickly
- After a few customers, delays are almost independent

#### Interpretation:

The system clears fast, so congestion does not last long.

## Why is this example **IMPORTANT** for simulation?

### Wrong assumption:

Simulation outputs are independent.

### Reality:

Simulation outputs are **correlated**, especially in queues.

Because of correlation:

- Standard confidence intervals may be **wrong**
- Variance estimates may be **too optimistic**

Optimistic” er mane হলো — অতিরিক্ত আশাবাদী বা বাস্তবের চেয়ে ভালো ধরে নেওয়া

## Example 4.23: Inventory Cost Correlations

For an (s,S) inventory system:

- $r_2$  is **positive** (ordering often happens every other month)
- $r_1$  is **negative**  
(a large cost one month usually means a small cost next month)

This shows that **simulation output can have both positive and negative correlations**.

## Warm-Up Period in Simulation

At the start of a simulation:

- The process is **usually not covariance-stationary**

After a **warm-up period**, the output:

- Becomes **approximately covariance-stationary**
- Only data **after the warm-up period** should be used for analysis

## Why IID Assumption Fails for Simulation Output

In real simulation output:

- Observations are **almost always correlated**
- They are **not IID**

### Danger

If we treat correlated simulation output as IID:

- Variance is underestimated (too optimistic)
- Confidence intervals become misleading
- Final conclusions may be wrong

## Link to Covariance-Stationary Model

To analyze correlated simulation data, we use the **covariance-stationary stochastic process model** (Section 4.3), which:

- Allows correlation between observations
- Gives a realistic framework for simulation output analysis

## Key Exam Points (Very Important)

- Sample mean is an **unbiased estimator** of population mean
- Sample variance is an **unbiased estimator** of population variance
- Larger sample size → smaller variance → better accuracy
- IID data → zero correlation
- Simulation output data are **usually correlated**
- Treating simulation data as IID is **dangerous**

Here's a **clear, exam-friendly summary** of Sections 4.6 and 4.7 from your text:

## 4.6 Strong Law of Large Numbers (SLLN)

### Statement (Theorem 4.2):

If  $(X_1, X_2, \dots, X_n)$  are IID random variables with finite mean  $(m)$ , then

$$\left[ \begin{array}{l} \bar{X}(n) \rightarrow m \quad \text{w.p. 1 as } n \rightarrow \infty \\ \end{array} \right]$$

### Meaning:

- As the number of observations  $(n)$  becomes very large, the **sample mean  $(\bar{X}(n))$**  almost surely gets **arbitrarily close to the true mean  $(m)$** .
- This is a **guarantee for convergence** rather than just a probability statement.
- It is “strong” because it applies **almost surely** (with probability 1) rather than just in expectation.

### Example 4.31:

- Suppose  $(X_i)$  are normal with mean  $(m = 1)$  and variance  $(s^2 = 0.01)$ .
- The plot of  $(\bar{X}(n))$  shows that for  $(n \geq 28)$ ,  $(\bar{X}(n))$  differs from  $(m)$  by **less than 1%**, illustrating SLLN.

### Key Point for Exams:

- SLLN ensures **accuracy of sample mean** when **sample size is large**.
- Often used to justify using averages from simulations or experiments.

## 4.7 Danger of Replacing a Distribution by Its Mean

### Problem:

- Sometimes, analysts replace a **random input** (distribution) with its **mean value** in simulation models.
- This is **dangerous** because it ignores the **variability** of the input.

### Example 4.32:

- M/M/1 queue with:
  - Exponential interarrival times, mean = 1 min
  - Exponential service times, mean = 0.99 min
- True average **queue delay** = 98.01 min

- If we replace distributions by their means (deterministic arrivals and service times), **queue delay becomes 0**, which is **completely wrong**.

**Lesson:**

- Both **mean and variance** of input distributions **affect simulation outputs**.
- Replacing a distribution by its mean **ignores randomness** and leads to misleading results.

**Exam Tip:**

- Strong law  $\rightarrow$  sample mean  $\rightarrow$  close to population mean for large  $n$
- Don't replace distributions by means  $\rightarrow$  variance matters for simulation accuracy